

Tutorial Clarity - Multi-Voice Bug Report for Claude Code

Quick Summary

A YouTube video tutorial translation app with multi-speaker voice support has a critical bug: voices play incorrectly despite correct detection, generation, and storage. Voices sometimes change mid-sentence.

The Flow That Should Work

1. User loads YouTube video with 3 speakers
2. AssemblyAI detects speakers A, B, C
3. System maps to speaker_0, speaker_1, speaker_2
4. User assigns voices: speaker_0=Nova, speaker_1=Onyx, speaker_2=Shimmer
5. System generates audio for each segment with correct voice
6. Scheduler plays audio synchronized with video
7. Each segment plays with its assigned voice

What Actually Happens

Steps 1-5 work perfectly (verified via logs). Step 6-7 broken: wrong voices play, sometimes changing mid-sentence.

Key Files to Review

1. `app/components/ClarifyAudioPanel.tsx` - Main audio generation & scheduling (~1800 lines)
2. `app/watch/page.tsx` - UI and configuration (~2400 lines)
3. `app/api/detect-speakers/route.ts` - AssemblyAI integration
4. `app/api/multi-voice-tts/route.ts` - OpenAI TTS API
5. `app/utils/matchSpeakerSegments.ts` - Speaker label matching
6. `VOICE_BUG_DIAGNOSTICS.md` - Detailed diagnostic report

The Mystery

- Generation logs show: "Generated segment 0 with voice=nova" ✓
- Cache stores: voice="nova" at index 0 ✓
- Post-generation audit: 3 different voices confirmed ✓
- But playback sounds wrong ✗

Key clue: Voices change mid-sentence → scheduler is switching audio blobs during playback

Suspected Issues

1. Scheduler fetching wrong audio blob index

2. Audio blobs stored at wrong indices in `cacheRef`
3. Segment indices not aligned with audio indices
4. Overlapping audio playback (multiple blobs playing simultaneously)

Architecture

ClarifyAudioPanel.tsx manages:

- `cacheRef`: `{ [segIndex]: { url, voice, generating } }` `audio blob` storage
- `frozenVoiceMapRef`: `{ speaker_0: "nova", ... }` voice assignments
- `assemblyAISpeakerMapRef`: `Map<segIndex, speakerId>` AssemblyAI labels
- `translatedTxRef`: `ClarifyTranscriptSegment[]` transcript segments
- `schedulerRef`: `setInterval` loop timing-based playback
- `audioRef`: `HTMLAudioElement` `current` playing audio

Playback flow:

- `schedulerRef` fires `every ~100ms` checks `currentTime` vs segment timestamps
- finds correct segment `index` fetches `cacheRef[index].url` plays audio

What We Need

Help identifying the bug in the audio playback/scheduling logic. All generation is confirmed working - the issue is purely in playback.

Project Details

- Framework: Next.js 14
- Audio: HTML5 Audio element + Object URLs from fetch blobs
- TTS: OpenAI API (nova/onyx/shimmer/echo/fable/alloy voices)
- Speaker Detection: AssemblyAI (file upload pipeline)
- Branch: `feature/v153-truly-clean`

Diagnostic Tools Available

- `window.testAudioBlobs()` — manually plays first 5 cached audio blobs to verify correct voices
- POST-GENERATION VOICE AUDIT in console logs — confirms voice distribution after generation
- Manual Detection button — gap-based speaker detection (no API needed)